

SAI TEJA SRIVILLIBHUTTURU

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4+ Years Software Engineering | 1+ Year ML/AI Research

EDUCATION

University of Texas at Arlington — M.S. Computer Science & Engineering Aug 2023 – May 2025

Relevant Coursework: **Neural Networks, Machine Learning, Computer Vision, Artificial Intelligence, Deep Learning**, Data Mining, Data Analysis & Modeling, Design & Analysis of Algorithms, Numerical Methods, Scalable Systems & Optimization

Andhra University — B.Tech. Computer Science Engineering Jun 2015 – Apr 2019

Relevant Coursework: Data Structures & Algorithms, **Object-Oriented Programming**, Database Management Systems, Computer Networks, **Artificial Intelligence**, Big Data Analytics, Data Mining, Cyber Security, Discrete Mathematics

SKILLS

LLM & Training: PyTorch, TensorFlow, Hugging Face, vLLM, LoRA/QLoRA, PEFT, CUDA, Distributed Training (DDP/NCCL), Quantization (GPTQ/AWQ), Speculative Decoding

Evaluation & Agents: RAGAS, BERTScore, HumanEval, Red-Teaming, LangChain, LangGraph, Multi-Agent Pipelines, RAG, Pinecone, Prompt Engineering

CV & Deep Learning: CNN, Supervised Learning, Image Augmentation, CLIP, PyTorch

Cloud & MLOps: AWS (SageMaker, EC2, S3, Lambda, ECR), Docker, Kafka, MLflow, CI/CD

Software Engineering: Python, Java, SQL, Spring Boot, FastAPI, REST APIs, Microservices

Simulation & RL: SUMO/TraCI, DDQN, Actor-Critic, Sionna 6G

EXPERIENCE

Qure.ai (Qure Technologies) | AI Solutions Engineer Intern Mar 2026 – May 2026

- Configured **clinical LLMs** for **radiology report parsing** and **EMR data extraction** across **EPIC/FHIR**-integrated hospital systems including Medstar, Mount Sinai, and UFL, supporting **6+ live health system sites**.
- Prepared **HIPAA**-aligned technical documentation including DPIAs, security questionnaires, and **EPIC bidirectional integration** guides for **clinical AI** deployment under real-time SLAs.

DentalScan (ReplyQuick AI LLC) | Machine Learning Engineer Intern Dec 2025 – Feb 2026

- Designed **CNN-based supervised ML** training pipelines on **AWS SageMaker**, S3, EC2, and Lambda for **intra-oral image classification** across 6 clinical categories on a **50K+ labeled dataset**.
- Improved **weighted F1** from **0.74 to 0.89** across 6 diagnostic categories by executing 15+ iterative **SageMaker** training runs with systematic per-category error analysis, **image augmentation**, and class-balancing strategies.
- Containerized **inference endpoints** via **Docker** and **REST APIs**; tracked all experiments using **MLflow** with version-controlled model registry and automated evaluation gates, compressing release cycles to same-day.
- Identified an 18% **minority-class recall** gap through **confusion-matrix** error analysis and rebuilt the **S3/Lambda augmentation pipeline** with weighted sampling and targeted synthetic generation.
- Scaled dataset ingestion to **50K+ images** using **AWS ECR**, SQS, Step Functions, and DynamoDB for metadata tracking, maintaining reproducible checkpoints and sustaining **0 regression incidents** across releases.

University of Texas at Arlington | Graduate Research Engineer Dec 2025 – Present

- Architected a deep **Stackelberg MARL** system in **SUMO** via **TraCI**, modeling congestion pricing as a two-level leader-follower problem across 3x3, 5x5, and 7x7 urban grids.
- Implemented **Dueling DDQN** and **A2C Actor-Critic** agents in **PyTorch** with replay buffers, gradient clipping, and entropy regularization, with toll actions physically modulating SUMO edge efforts per step.
- Quantified efficiency-equity tradeoffs using 6 **fairness metrics** (Gini, Theil, Atkinson) across 4 vehicle classes with calibrated VoT weights over 3 seeds with **95% CI**.
- Modeled a **digital twin** with Gaussian sensor noise, observation delays, and BPR-calibrated travel-time functions, validating robustness across demand variance from 0.05 to 0.5.
- Benchmarked **DDQN** and **Actor-Critic** against no-toll, marginal-cost, and static-toll baselines, producing 32 statistical figures including convergence curves and fairness radar charts.

- Developed **DTMAP**, a multi-objective **path planning** framework integrating **wireless digital twins** with a fine-tuned **LLM (GPT-4o-mini)** for **connectivity-aware navigation** in **6G/XR** environments; accepted at **IEEE OJCOMS 2026**.
- Designed a tunable scalarization parameter α enabling continuous **signal-distance trade-offs** across 21 values without retraining; achieved **1.9% outage probability** vs. 2.3% for **RL baseline**.
- Built deterministic **path sanitization pipeline** raising raw **LLM structural validity from 65% to 100%**; benchmarked against **A***, greedy, **Q-learning**, **LLaMA 3.1/3.3-70B**, and **Qwen-2.5-72B** at **312 ms end-to-end inference latency**.
- Generated **3,150 training samples** across 630 source-destination pairs using **NVIDIA Sionna** ray tracing on **3D urban models** built with **OpenStreetMap** and **Blender**.

- Developed **CTMap**, an **LLM-enabled path planning** system fine-tuned on **10K+ Dijkstra-generated** routing samples using **OpenStreetMap** and **Sionna 6G** channel simulation; accepted at **IEEE ICC 2026 (CQRM Workshop)**.
- Addressed 31 peer-reviewer concerns including **outage probability analysis** and **alpha-mapping sensitivity studies**, directly contributing to **IEEE OJCOMS** acceptance.
- Supported graduate-level **Numerical Methods** course for 50+ students, assisting with algorithmic problem-solving, optimization, and computational modeling.

- Designed a **microservices-based** middleware platform using **API Gateway**, Circuit Breaker patterns in **Java** and **Spring Boot**, connecting 5+ distributed financial systems via **REST/SOAP APIs** handling **10K+ transactions**.
- Reduced manual processing overhead by **40%** by engineering automated **ETL pipelines** in **Python** and **Spark** processing **50K+ records** at 99.8% data integrity.
- Optimized high-volume **SQL** queries and **Spark** jobs improving throughput by **35%**; enforced **TDD** with 85%+ JUnit/Mockito coverage and **CI/CD** via GitHub Actions, cutting defects by **25%**.
- Mentored 3 junior engineers on Saga, Circuit Breaker, and TDD; led sprint planning for a 5-engineer team, delivering 2 consecutive milestones **2 weeks ahead of schedule**.

PROJECTS

ML / Computer Vision

- **Multi-Object Tracking** (SORT/DeepSORT, MOTA/MOTP, PyTorch, TraCI): Reproduced and extended **SORT/DeepSORT** tracking pipelines for autonomous vehicle perception, benchmarking **MOTA/MOTP** across occlusion scenarios with **supervised learning** baselines.
- **Foundation Model Fine-Tuning** (LLaMA, Mistral, LoRA, QLoRA, PEFT): Ran systematic **fine-tuning** experiments across **LoRA**, **QLoRA**, and full fine-tuning on **LLaMA** and **Mistral**, capturing data efficiency and loss convergence tradeoffs.
- **Amazon Hybrid Recommender** (Collaborative Filtering, ETL, NDCG, A/B Testing): Built a hybrid **collaborative filtering** recommender with **ETL pipelines**, evaluated via **NDCG**, **MRR**, and **A/B testing**.

LLM / AI Systems

- **Distributed LLM Pre-Training** (PyTorch DDP, NCCL, CUDA): Built a production-grade **distributed training** system achieving **3.50x speedup** on 4 GPUs at 87.5% parallel efficiency, processing **152K tokens/second** with fault-tolerant checkpointing and P95 dashboards.
- **vLLM Throughput Benchmark** (vLLM, ONNX, Speculative Decoding, GPTQ/AWQ): Achieved **18.6x throughput gains** over Hugging Face via **speculative decoding** and **GPTQ/AWQ quantization** at sub-100ms P95 latency.
- **Multi-Agent Research Assistant** (LangGraph, LangChain, ChromaDB, FastAPI, RAGAS): Implemented a 4-stage **multi-agent pipeline** (Researcher, Critic, Synthesizer, Evaluator) with **RAGAS** evaluation, **ChromaDB RAG** retrieval, **FastAPI** backend, and Gradio frontend.
- **LLM Code-Agent Eval Benchmark** (Grok, Gemini, HumanEval, BERTScore): Developed sandboxed evaluation infrastructure measuring **pass@1** and **BERTScore** across 164 **HumanEval** tasks and 3 models with **ROUGE**-based output quality scoring.
- **Red-Teaming & Safety Evaluation** (LLMs, Diffusion Models, Ablation Studies): Conducted structured **red-teaming** surfacing jailbreaks, hallucinations, and adversarial vulnerabilities across **LLMs** and diffusion models, producing ablation-backed risk-tiered failure taxonomies.
- **TeluguGPT** (GPT, SFT, Low-Resource NLP): Adapted a **generative language model** on Telugu corpus using **SFT** for 80M+ speakers, addressing tokenization, script normalization, and cultural context challenges on Linux-based GPU environments.

PUBLICATIONS

- Md. Salik Parwez et al., "Digital Twin-Guided AI Path Planning for Connectivity-Aware Mobility," **IEEE Open Journal of the Communications Society (OJCOMS)**, Accepted April 2026.

- Md. Salik Parwez et al., "CTMap: LLM-Enabled Connectivity-Aware Path Planning in Millimeter-Wave Digital Twin Networks," **IEEE ICC 2026 (CQRM Workshop)**, Accepted January 2026.

CERTIFICATIONS

Advanced Large Language Model Agents — UC Berkeley EECS | **AWS Certified Data Engineer – Associate** | **Microsoft Certified: Fabric Data Engineer Associate** | AI Evals for Everyone | **Oracle GenAI Professional** | Oracle AI Vector Search Certified Professional | Salesforce Agentforce Specialist | Salesforce Certified AI Associate